



IEMS 5709 Emerging Topics in Information Engineering

Practical Lecture 1: Nvidia Orin NX & Jetson AI Lab

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Demonstrator: (TA) Alex WANG

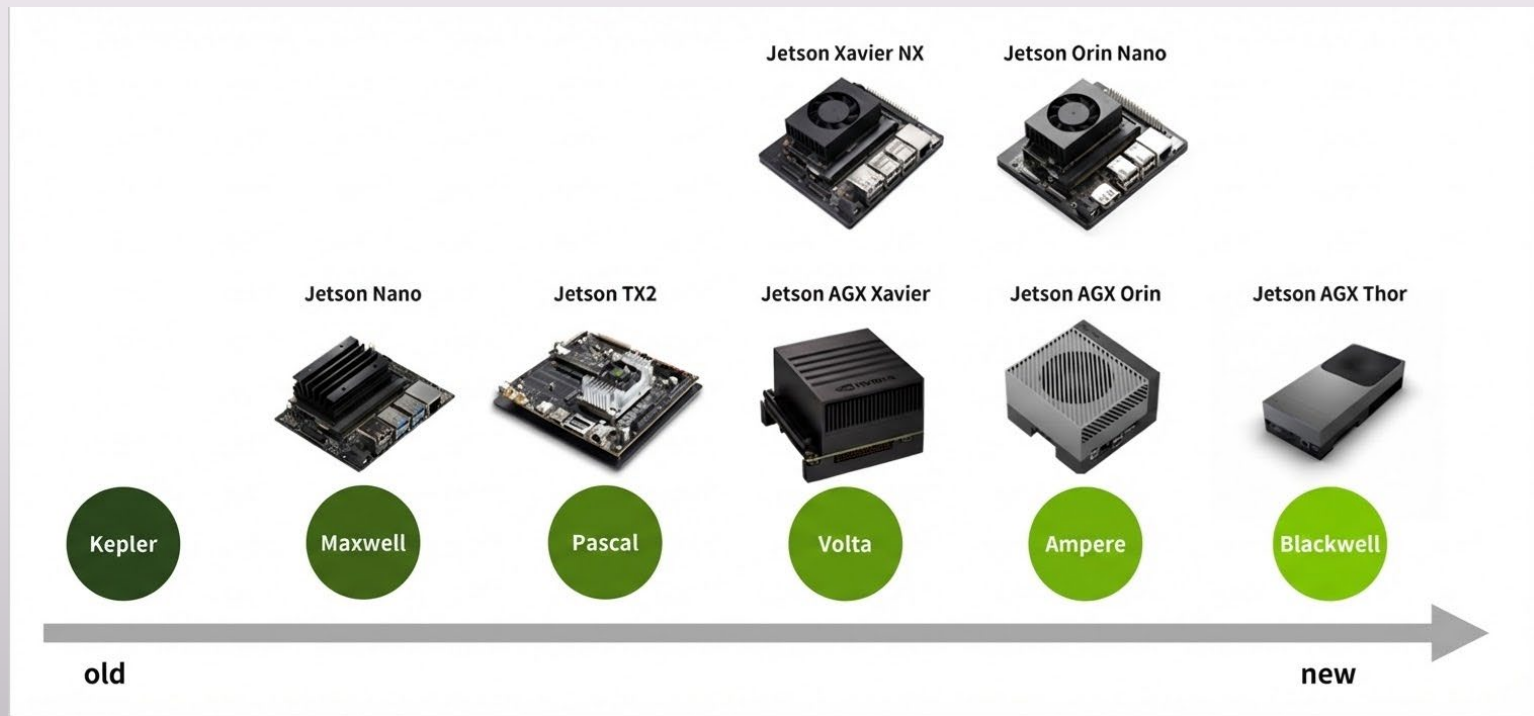
2026/02/07



NVidia Orin NX Hardware

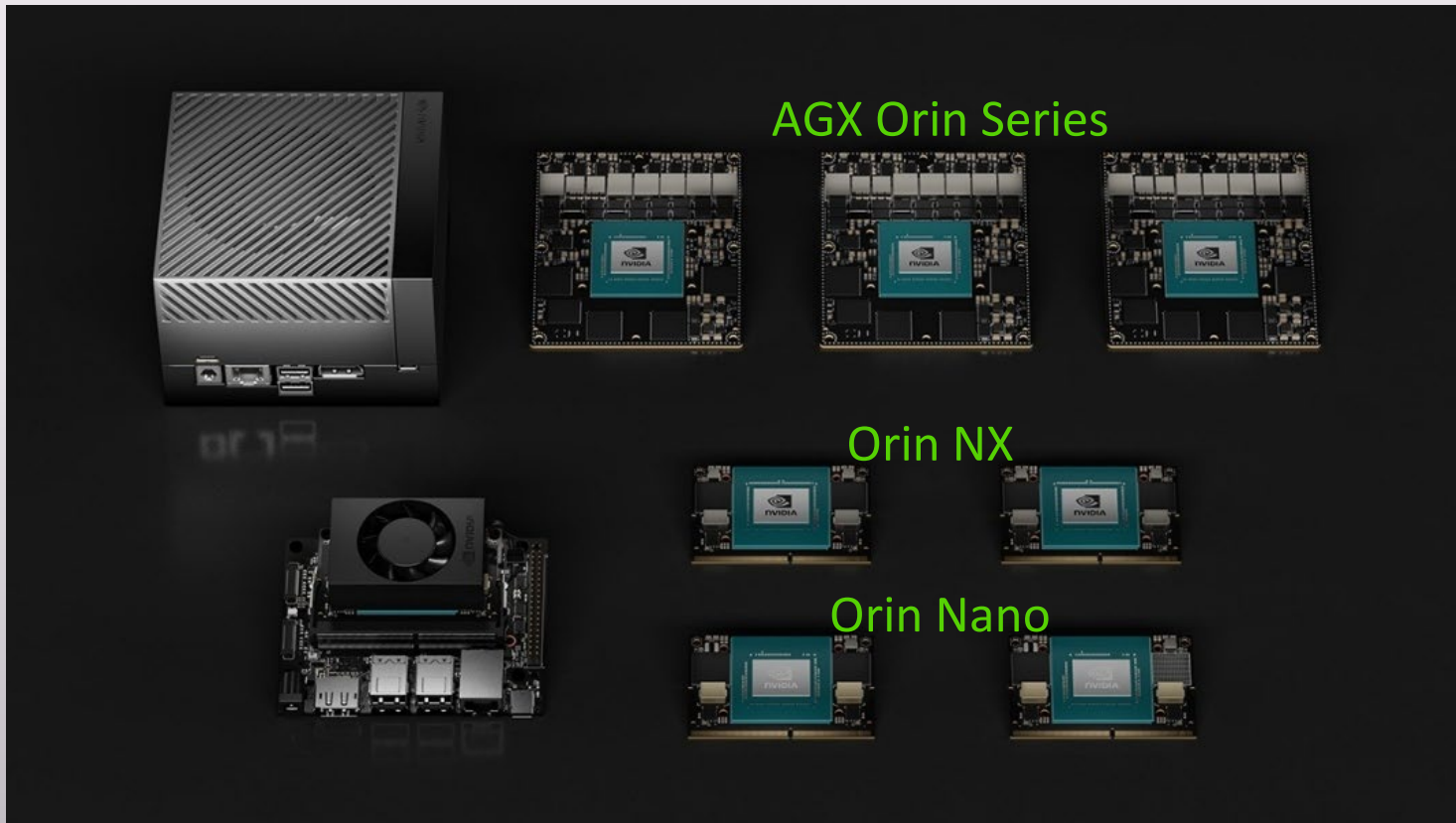


Evolution of Nvidia Jetson



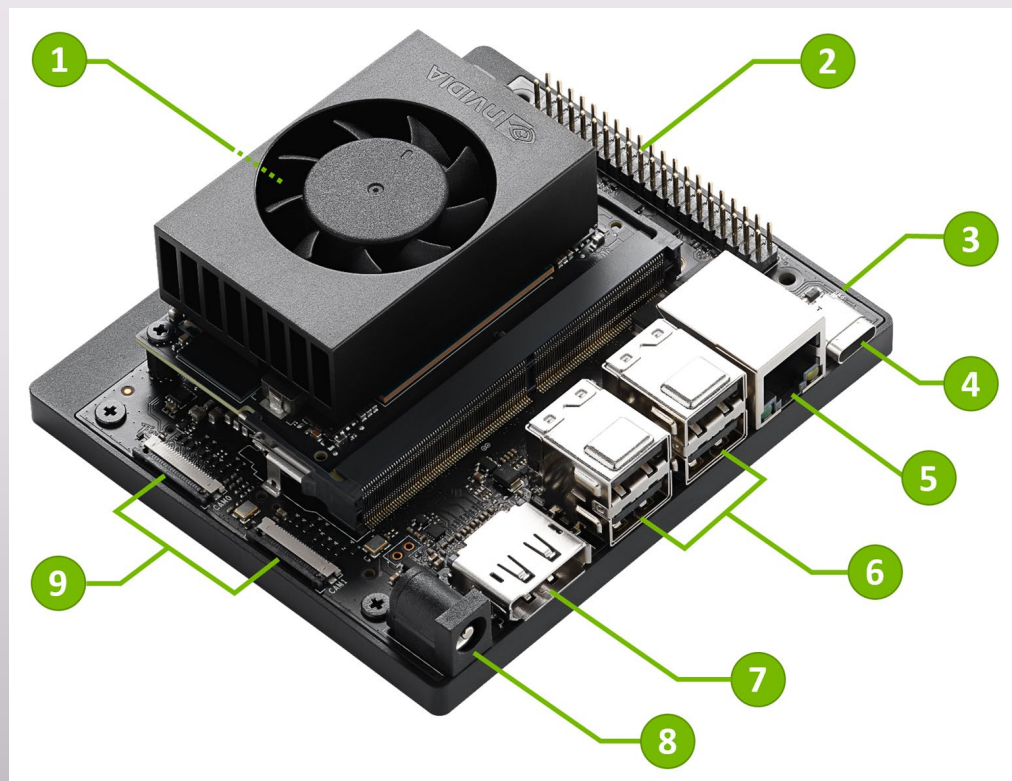


Nvidia Jetson Orin Lineup





Jetson Orin NX Developer Kit

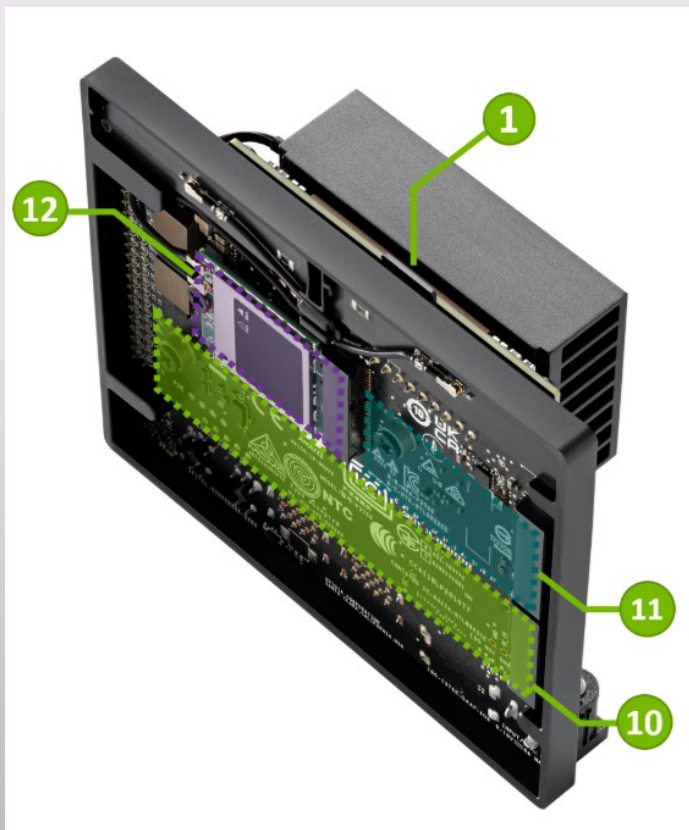


- 1 microSD card slot for main storage
- 2 40-pin expansion header
- 3 Power indicator LED
- 4 USB-C port for data only
- 5 Gigabit Ethernet port

- 6 USB 3.1 Type A ports (x4)
- 7 DisplayPort connector
- 8 DC Barrel jack for 19V power input
- 9 MIPI CSI camera connectors



Jetson Orin NX Developer Kit



10	M.2 Slot (Key-M, Type 2280)	PCIe 3.0 x4
11	M.2 Slot (Key-M, Type 2230)	PCIe 3.0 x2
12	M.2 Slot (Key-E, Type 2230)	(populated)



Jetson Orin NX (Super)

New upgrade
greatly improved computing power

The computing power of the Jetson Orin NX SUPER developer kit is 67% higher than the previous generation, the GPU frequency is increased by 53%, helping developers meet the needs of large-scale AI projects.

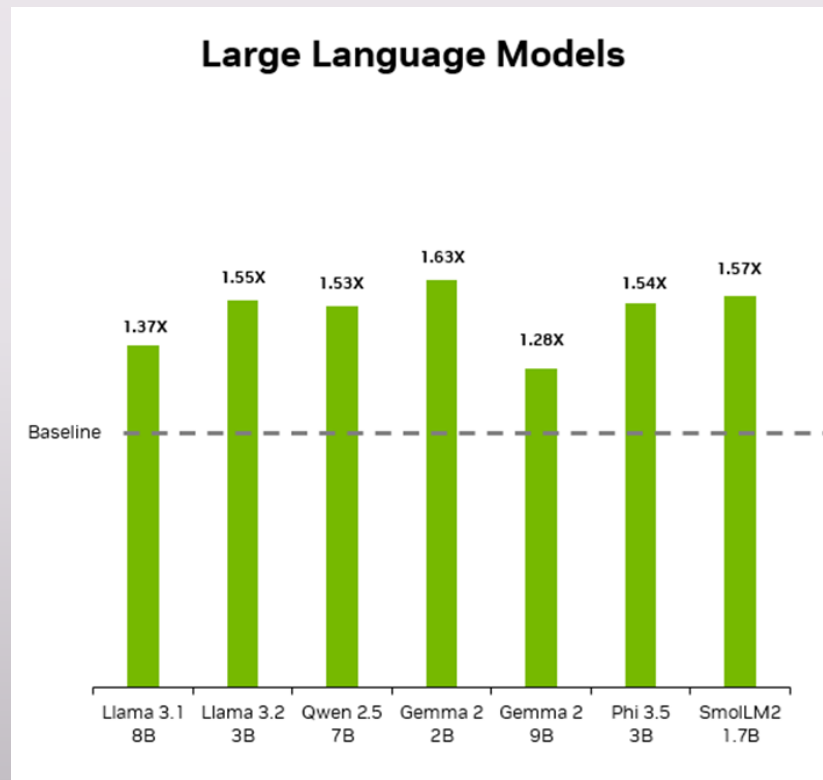


157
Tops

NVIDIA

Jetson Orin NX **SUPER Developer Kit**

- 8G / 16G RAM for choice
- 117 / 157-TOPS computing power
- LPDDR5 memory





Jetson Orin NX (Super)

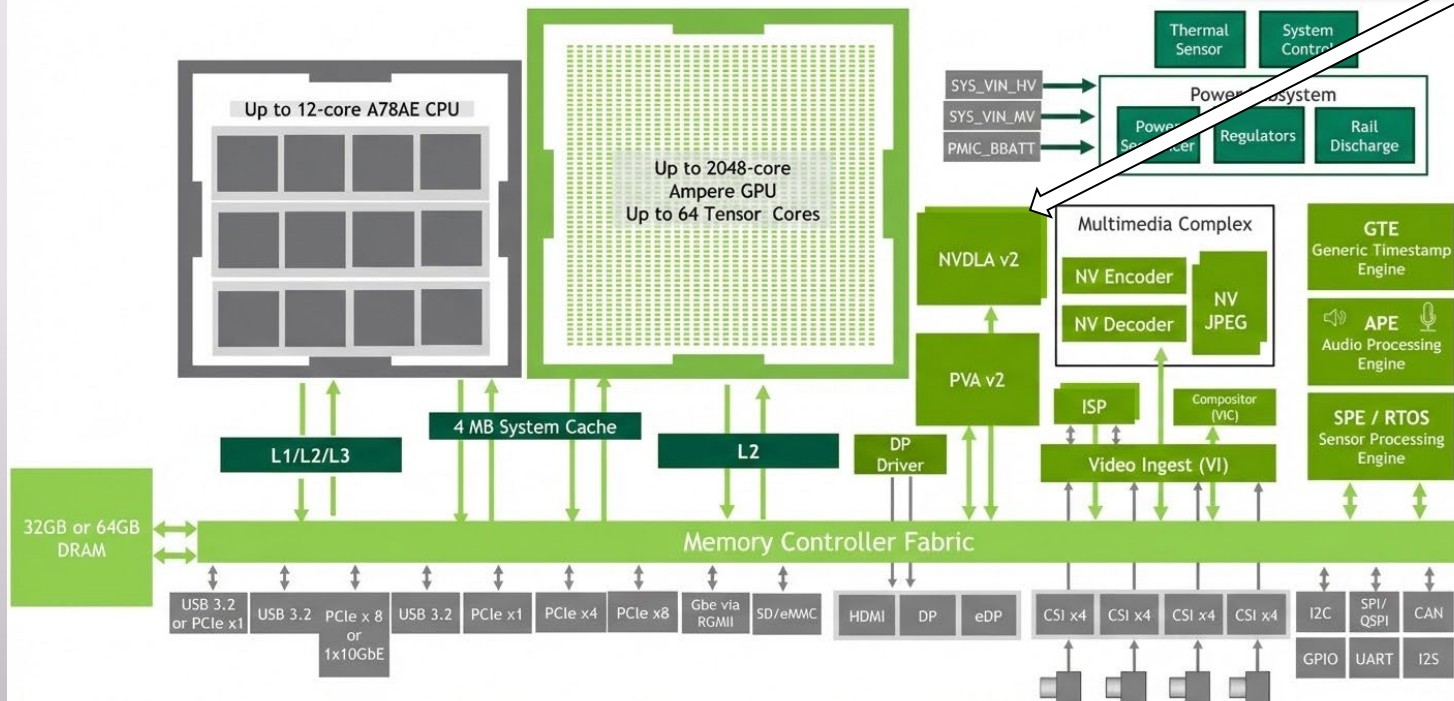
Feature	Orin NX 8GB	Orin NX 8GB (SUPER)	Orin NX 16GB	Orin NX 16GB (SUPER)
AI Performance (Sparse)	70 TOPS	117 TOPS	100 TOPS	157 TOPS
GPU Max Frequency	765 MHz	1173 MHz	918 MHz	1173 MHz
CUDA Cores	1024 (Ampere)	1024 (Ampere)	1024 (Ampere)	1024 (Ampere)
Tensor Cores	32	32	32	32
CPU Configuration	6-core Arm® Cortex®-A78AE	6-core Arm® Cortex®-A78AE	8-core Arm® Cortex®-A78AE	8-core Arm® Cortex®-A78AE
DL Accelerator	1x NVDLA v2	1x NVDLA v2	2x NVDLA v2	2x NVDLA v2
Memory (RAM)	8GB LPDDR5	8GB LPDDR5	16GB LPDDR5	16GB LPDDR5
Power Budget	10W 15W 20W	10W 15W 25W 40W	10W 15W 25W	10W 15W 25W 40W



Jetson Orin Block Diagram

Jetson AGX Orin Series Functional Block Diagram

NVDLA 2.0 (Deep Learning Accelerator)





NVDLA—Efficient Inference Engine

Feature / Specification	Jetson Orin NX Super (16GB)	RTX 3050 (Desktop)	RTX 3050 (Laptop 75W)
GPU Architecture	Ampere (GA10B)	Ampere (GA106)	Ampere (GA107)
Tensor Cores (3rd Gen)	32	80	64
CUDA Cores	1024	2560	2048
GPU Max Frequency	1173 MHz	~1777 MHz (Boost)	~1500 MHz (Boost)
INT8 Dense Compute	78.5 TOPS (GPU + DLA)	73 TOPS (GPU only)	~90 TOPS (GPU only)
BF16 Dense Compute	~19.2 TFLOPS (GPU only)	36.5 TFLOPS	~45 TFLOPS
Memory / VRAM	16GB (Unified LPDDR5)	8GB (GDDR6)	4GB / 6GB (GDDR6)
Memory Bandwidth	102.4 GB/s	224 GB/s	192 GB/s
Power (TDP)	10W - 40W	130W	35W - 80W



Nvidia Jetson AI Lab



Jetson AI Lab: A Collection of Resources

NVIDIA Jetson AI Lab

[Home](#) [Models](#) [Tutorials](#) [Benchmarks](#) [Projects](#) [Research Group](#)

Generative AI at the Edge

Bring generative AI to the world with NVIDIA[®] Jetson[™]

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An illustration of a black and silver Jetson module on a green base. Above it are two green speech bubbles; the left one contains three horizontal lines, and the right one contains a tiger's face.

You can find **Models, Tutorials, Benchmarks, and Community Projects**



Jetson AI Lab Models

All Models

Compact directory, grouped by model family

Google Gemma3

FunctionGemma	Run	Details
Gemma 3 270M	Run	Details
Gemma 3 1B	Run	Details
Gemma 3 4B	Run	Details
Gemma 3 12B	Run	Details
Gemma 3 27B	Run	Details

Meta Llama 3

Llama 3.2 3B	Run	Details
Llama 3.1 8B	Run	Details
Llama 3.1 70B	Run	Details

Mistral AI Ministral 3

Ministral 3 3B Instruct	Run	Details
Ministral 3 8B Instruct	Run	Details
Ministral 3 14B Instruct	Run	Details
Ministral 3 3B Reasoning	Run	Details
Ministral 3 8B Reasoning	Run	Details
Ministral 3 14B Reasoning	Run	Details

NVIDIA Nemotron

Nemotron3 Nano 30B-A3E	Run	Details
Nemotron Nano 9B v2	Run	Details
Nemotron Nano 12B VL	Run	Details

Qwen Qwen3

Qwen3 4B	Run	Details
Qwen3 8B	Run	Details
Qwen3 30B-A3B (MoE)	Run	Details
Qwen3 32B	Run	Details
Qwen3 VL 4B	Run	Details
Qwen3 VL 8B	Run	Details

W4 (Weights = 4-bit)
A16 (Activations = 16-bit)

Text

Qwen3 4B

Alibaba's efficient 4 billion parameter instruction-tuned language model

[Run on Jetson](#)

Memory Requirement

4GB RAM

Precision

W4A16

Size

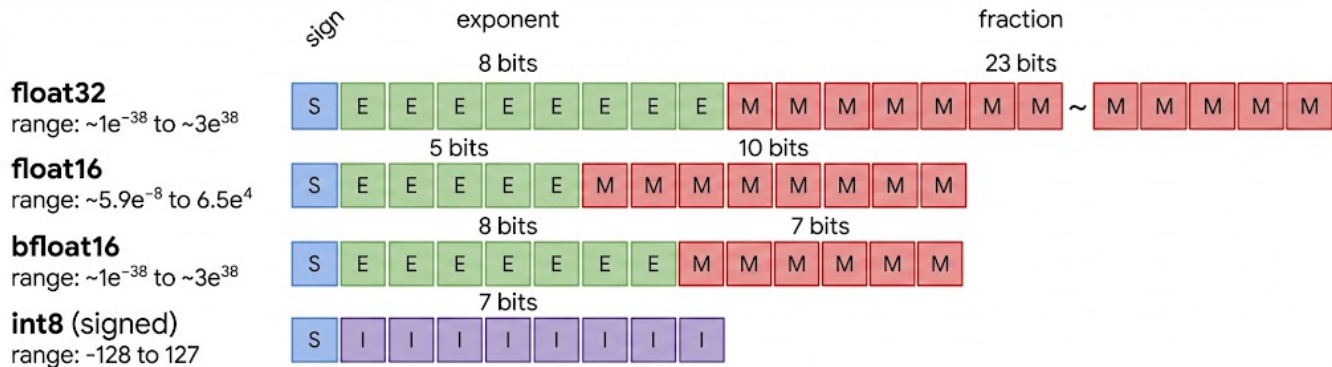
2.5GB



Model Weights Precision

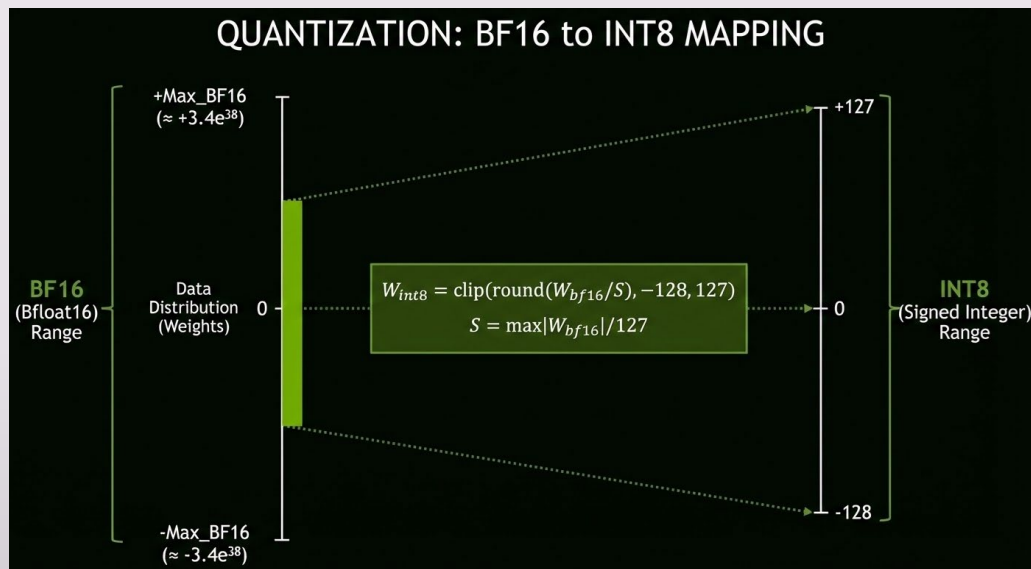


NVDLA 2.0 supports **FP16/INT8**, but most models are trained in **BF16**.





Model Weights Quantization

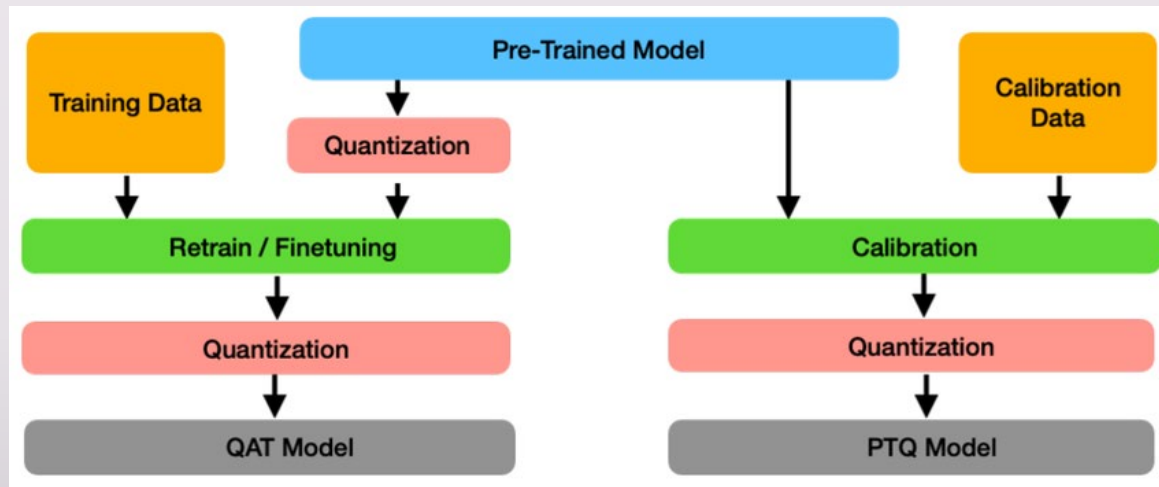


Simple Quantization Illustrated:

Mapping the **vast dynamic range** of BF16 (Left) into the **compact bucket** of INT8 (Right) using a single parameter: the Scale Factor.



Quantization: PTQ VS QAT

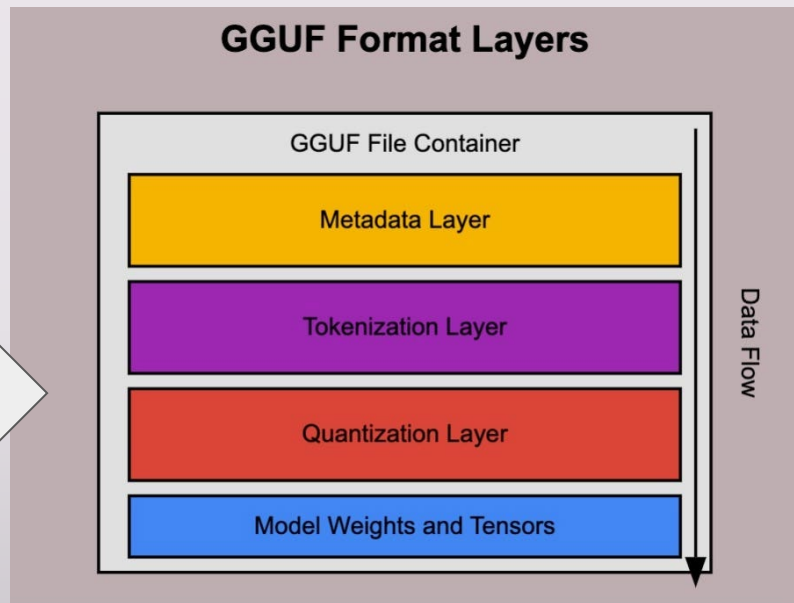
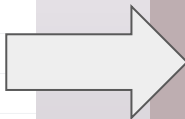


QAT (Quantization-Aware Training) models typically perform better than PTQ (Post-Training Quantization) models. Some model providers offer QAT checkpoints (for example, Google's Gemma 3 series), but most do not.



GGUF Format: All-in-One Model Packaging

main	Qwen3-0.6B	1.52 GB	Go to f
littlebird13	Create LICENSE	c1899de	VERIFIED
.gitattributes		1.57 kB	↓
LICENSE		11.3 kB	↓
README.md		14 kB	↓
config.json	1. Model Config File	726 Bytes	↓
generation_config.json	2. Generation Config File	239 Bytes	↓
merges.txt	3. Tokenizer Config File	1.67 MB	↓
model.safetensors	4. Tokenizer Vocab Files (vocab.json + merges.txt)	1.5 GB xet	↓
tokenizer.json	5. Tokenizer	11.4 MB xet	↓
tokenizer_config.json		9.73 kB	↓
vocab.json	6. Model Weights	2.78 MB	↓



Qwen3-0.6B-BF16.gguf 1.2 GB xet ↓



LLMs & SLMs



ChatGPT



Claude



Gemini

LLMs are massive models, typically too large for direct edge deployment.

SLMs are smaller models designed for resource-constrained edge environments.

Quantization (like INT8/INT4) is essential to fit SLMs on edge hardware like Orin NX.

Open weights



LLaMA



deepseek



Mistral AI



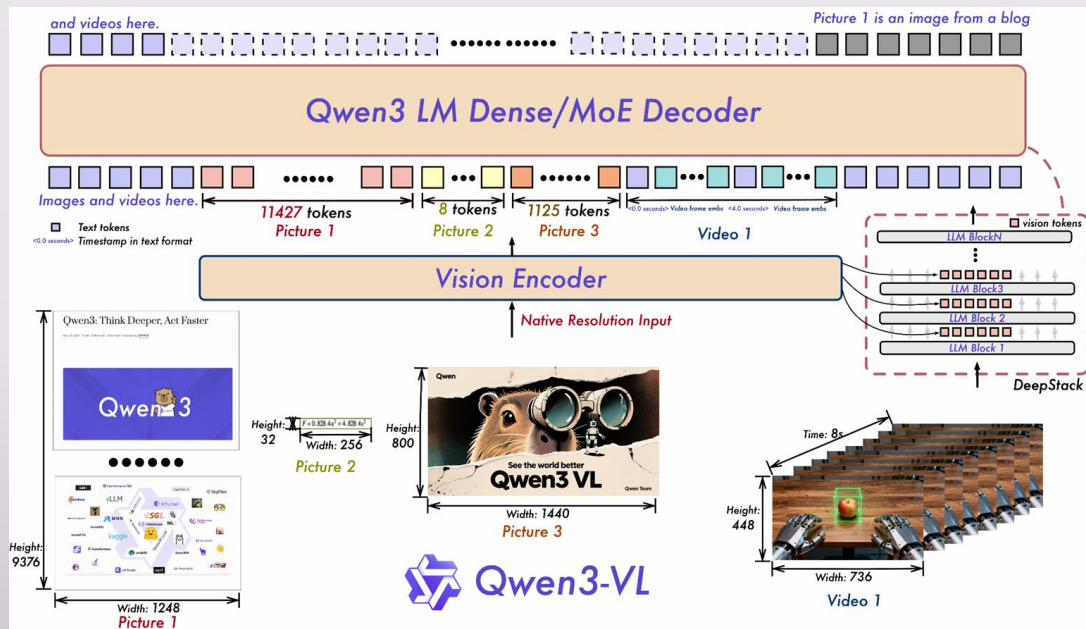
Gemma



Qwen



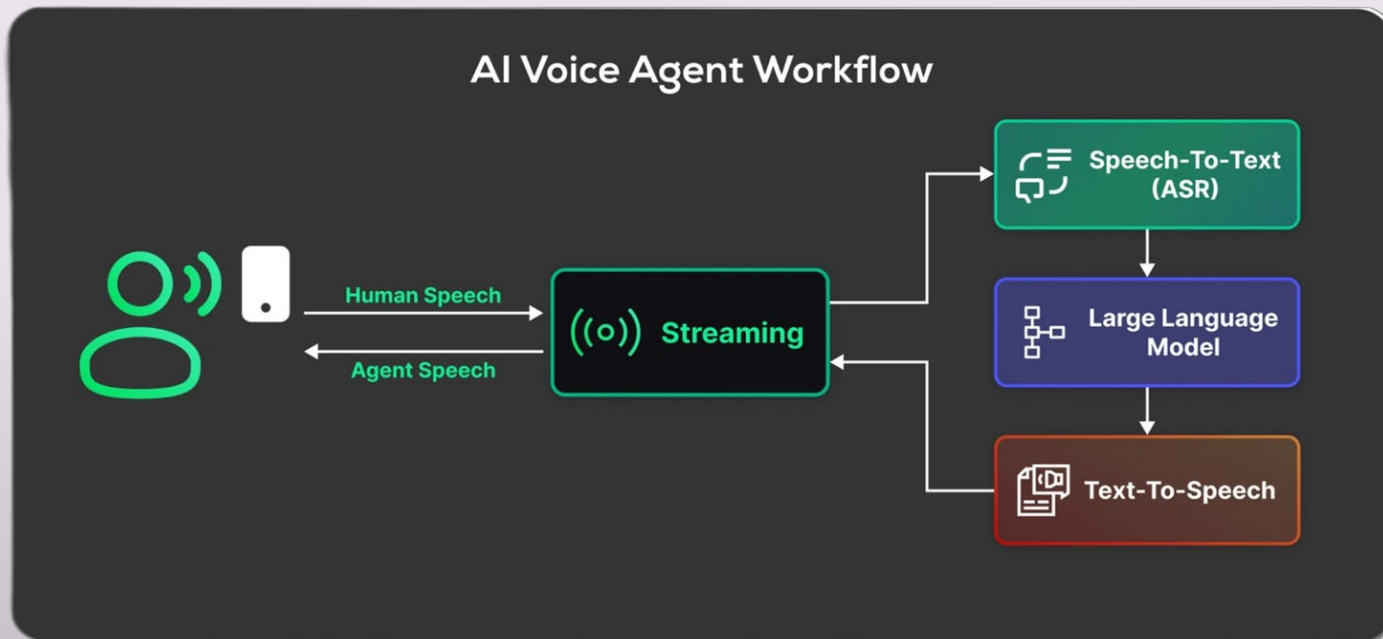
VLMs (Vision Language Models)



VLMs support understanding, perceiving, and reasoning about visual content of images and even videos.



Voice AI Pipeline





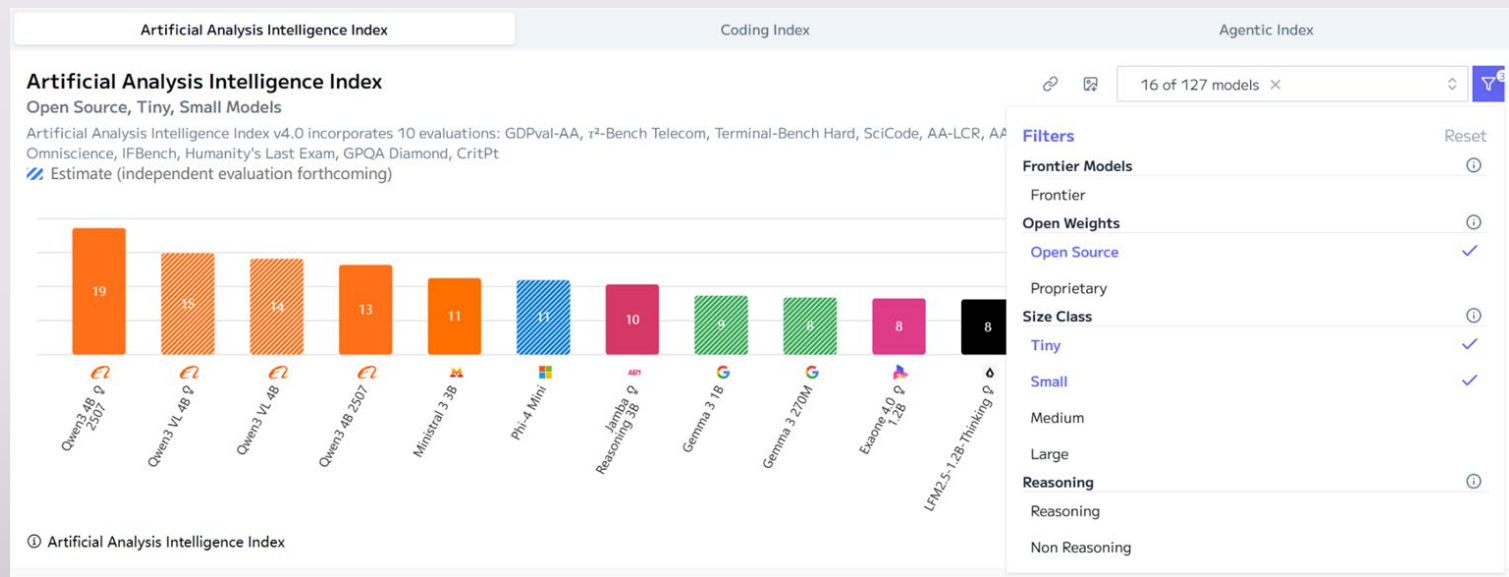
Voice AI Models

Component	Role	Input	Output	Typical Model Size (Edge)	Typical Latency
VAD (Voice Activity Detection)	The Gatekeeper Filters silence/noise to save compute.	Raw Audio (<i>PCM Stream</i>)	Boolean Flag / Time-stamps	< 1 MB – 50 MB (e.g., <i>Silero</i>)	< 10 ms – 30 ms (<i>Negligible</i>)
ASR (Automatic Speech Recognition)	The Ears Converts speech to text.	Speech Waveform	Text String (<i>Transcription</i>)	75 MB – 1.5 GB (e.g., <i>Whisper-small</i>)	200 ms – 800 ms (<i>Goal: RTF < 0.2</i>)
LLM (Large Language Model)	The Brain Processes intent & generates reply.	Text Prompt (<i>History + Query</i>)	Text Stream (<i>Token by token</i>)	3 GB – 15 GB (<i>INT8 Quantized</i>) (e.g., <i>Qwen-3-8B</i>)	TTFT: 300 – 800 ms (<i>Time To First Token</i>)
TTS (Text-to-Speech)	The Mouth Synthesizes audio response.	Text Stream	Audio Waveform	50 MB – 500 MB (e.g., <i>CosyVoice2</i>)	100 ms – 400 ms (<i>Time to First Audio</i>)

The "1.5-Second Rule": To achieve a natural, conversational feel (similar to a human pause), the sum of these latencies needs to be minimized.



How to Find Suitable Models?



You can find SOTA models for your task on the [Artificial Analysis](#) website. Remember to set the filter to 'Open Source' and limit the 'Size Class'.



How to Find Suitable Models?

The screenshot displays the Hugging Face Models interface. On the left, there are filters for Tasks (Text Generation, Any-to-Any, Image-Text-to-Text, Image-to-Text, Image-to-Image, Text-to-Image, Text-to-Video, Text-to-Speech, +44), Parameters (a slider from <1B to >500B), Libraries (PyTorch, TensorFlow, JAX, Transformers, Diffusers, sentence-transformers, Safetensors, ONNX, GGUF, Transformers.js, MLX, +41), and Apps (vLLM, llama.cpp, MLX LM, LM Studio, Ollama, Jan, Draw Things, +7). The main area shows a list of models with details like name, task, update time, downloads, and likes.

Model Name	Task	Updated	Downloads	Likes
zai-org/GLM-OCR	Image-to-Text	Updated 3 days ago	149k	694
openbmb/MiniCPM-o-4_5	Any-to-Any	Updated about 5 hours ago	2.39k	542
stepfun-ai/Step-3.5-Flash	Text Generation	Updated about 21 hours ago	10.9k	455
ACE-Step/Ace-Step1.5	Text-to-Audio	Updated 3 days ago	16.2k	387
Qwen/Qwen3-ASR-1.7B	Automatic Speech Recognition	Updated 7 days ago	132k	387
robbyant/lingbot-world-base-cam	Image-to-Video	Updated 4 days ago	281	
Tongyi-MAI/Z-Image	Text-to-Image	Updated 9 days ago	14k	849

Pull the weights of your chosen model from [Huggingface](https://huggingface.co).



Jetson AI Lab Benchmarks

Performance Comparison

Benchmarks on Jetson running vLLM

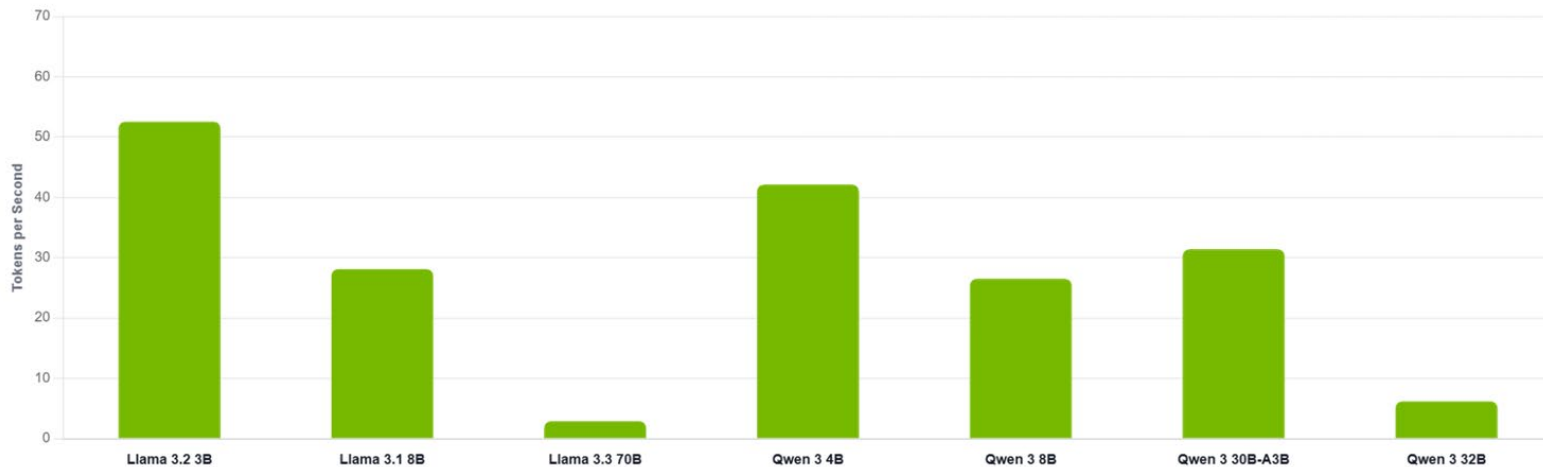
Tokens/s (Concurrency 1) on Jetson Orin

Orin

Thor

Tokens/s (c=1)

Tokens/s (c=8)





Jetson AI Lab Tutorials



Text Generation

Run LLM-based chat bot on Jetson

[Learn More >](#)



Text + Vision

Run multimodal Vision-Language models to give your AI access to vision

[Learn More >](#)



Image Generation

Run diffusion models to generate stunning images interactively on Jetson

[Learn More >](#)



Distillation

Learn a technique to bring the power of a large foundation model to Jetson

[Learn More >](#)



Vision Transformer (ViT)

Vision Transformer examples, with models optimized to run in realtime on Jetson

[Learn More >](#)



NanoDB

Multimodal vector database that uses embeddings for txt2img and img2img similarity search

[Learn More >](#)



Whisper

Run OpenAI's Whisper model to transcribe audio

[Learn More >](#)



Llamaspeak

Live interaction with Llama LLM using NVIDIA Riva AST/TTS

[Learn More >](#)



Jetson AI Lab Community Projects

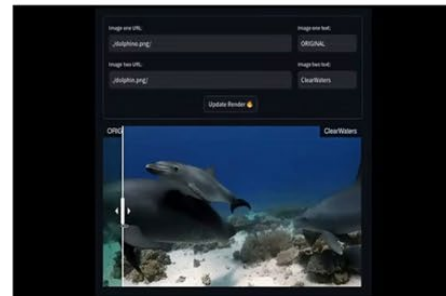


Armoury - Bringing GLaDOS to Life with Robotics and AI

Dave's

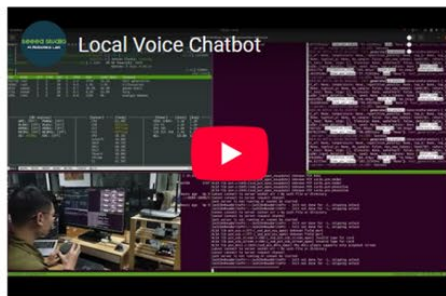


Nurgaliyev Shakhizat - Vision2Audio: Blind Assistant Device



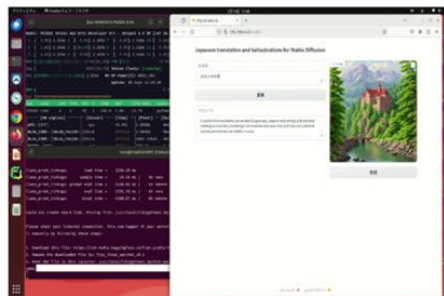
Pham - ClearWater: Underwater Image Enhancement with Generative AI

Vy



Studio - Local Voice Chatbot: Deploy Riva and Llama2 on Jetson

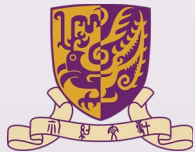
Seeed



Toshihiko Aoki - Japanese NMT Translation for Stable Diffusion



NVIDIA Webinar - Bringing Generative AI to Life with NVIDIA Jetson



Reference

- [Jetson AI Lab | Jetson AI Lab](#)
- [Jetson AGX Orin for Next-Gen Robotics | NVIDIA](#)
- [Jetson Orin Nano Developer Kit Getting Started Guide | NVIDIA Developer](#)
- [Comparison of AI Models across Intelligence, Performance, Price | Artificial Analysis](#)
- [Models – Hugging Face](#)
- [Models - Ollama](#)